

NICHOLAS B. LUDWIG

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SUMMARY

Scientist-turned-software engineer building agent frameworks, developer platforms, and simulation/data pipelines. PhD physical chemist with production experience in Rust, Python, TypeScript, and Go for diverse projects including Wasm runtimes, MCP-based agents, and HPC scientific computing. Experience turning ambiguous requirements into concrete tooling to unlock value for researchers and devs.

EXPERIENCE

Hyperware FKA Kinode FKA Uqbar

February 2022 - Present

Software Developer to Lead Developer

- Architected and built spider, an MCP client/server agent framework with tool loops, conversation storage, and reliable orchestration across Hyperware process APIs
- Designed kibitz, a coding agent that iteratively writes, tests, and revises code using MCP tools
- Developed document-driven AI coding workflow using Codex with custom async agent supervision
- Built developer platform (hyperdrive) focused on fault-tolerance and concurrent processes with message-passing and sandboxing for Wasm userspace; adapted bleeding-edge libraries
- Designed and wrote CLI developer tooling for Hyperware, including templates, building, managing real and “fake” nodes (for development), testing, and deployment capabilities (kit)
- Shipped 10+ full-stack applications on Hyperware, from idea to working prototype in under a week, including a real-time multi-user p2p voice app over WebSocket
- Partnered with stakeholders to ship smart contracts within regulatory and technical constraints

Mod9 Technologies DBA Remeeting

May 2020 - February 2022

Backend Engineer

- Wrote Python REST API and asyncio SDK: simple, dev-friendly interfaces for ASR
- Improved maintainability of existing code and significantly reduced cost of infrastructure

University of Chicago

January 2015 - August 2019

Graduate Researcher

- Optimized simulator by trading memory for speed, reducing redundant computation: cached neighbors and lattice distance factors, used Ewald summation, and computed only the energy change each step
- Designed Python/Bash pipeline for running HPC jobs and analyzing results on Slurm: produced and post-processed tens to hundreds of CPU-years of data
- Resolved ambiguous scientific questions via simulation; co-authored 4 papers including one in *Nature*

EDUCATION

University of Chicago

Ph.D., Physical Chemistry

December 2019

University of California, Merced

B.S., Chemical Sciences, with honors; Minor in Sociology

May 2014

SKILLS

Languages	Rust, Python, TypeScript, C, Go, Bash
AI/Agents	MCP, agent frameworks, Codex, Claude Code, tool use, LLM workflows
Systems/Data	perf, FlameGraph, Valgrind, Wasm, Tokio, Docker, WebSocket, OpenMP
Interpersonal	Academic & technical writing, presentations, prioritization, project management